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Module Handbook

OCOM5102M Algorithms 22/23 F02

Contents

Module Handbook 1
Contents 1
Accessibility 2
Welcome to the Module 3
Module Team 3
Module Objectives 3
Learning Outcomes 3
Syllabus 5
Unit 1 – Fundamentals of algorithms and algorithm analysis 5
Unit 2 – Graphs and algorithm design principles 5
Unit 3 – Algorithm design principles 6
Unit 4 – Important data structures and introduction to complexity 6
Unit 5 – Reductions and branch and bound 6
Unit 6 – Dealing with NP-hard problems 6
Key Dates 8
Module Assessment 9
Assessment 1: Coursework 1 9
Assessment 2: Coursework 2 12
Assessment 3: Final Assessment 15

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Reading List 18

Additional Information and Support 18

Key Documents 18

Skills@Library 18

General Module Forum 18

Student Success Advisor 18

Recording Policy 19

Complaints 19

Accessibility

For information in alternative formats please email DESstudents@leeds.ac.uk or speak with your Student Success Advisor.

Welcome to the Module

Welcome to OCOM5102M Algorithms.

This handbook contains administrative information related to the module to help you to organise study time for the module. We hope that you find this material useful and the work you do on the module rewarding.

Module Team

Your module leader will be Sarah Fores. You will have the opportunity to speak to them during the live weekly webinar sessions. If you need to contact the module leaders directly, you can find their details in the ‘Staff Information’ section of Minerva. The module was designed by Dr Sebastian Ordyniak and Dr Isolde Adler.

Module Objectives

Algorithms and algorithmic problem solving are at the heart of computer science. This module introduces students to the design and analysis of efficient algorithms and data structures. Students learn how to quantify the efficiency of an algorithm and what algorithmic solutions are efficient. Techniques for designing efficient algorithms are taught, including efficient data structures, standard methods such as Divide-and-Conquer and Dynamic Programming as well as more advanced techniques for computationally intractable problems and large data sets. This is done using illustrative and fundamental problems relevant to AI.

Learning Outcomes

Successful completion of this module will help you to:

- Demonstrate an understanding of what constitutes an efficient and an inefficient solution to a computational problem
- Analyse the efficiency of algorithms
- Evaluate and justify appropriate ways to provide efficient solutions for computational problems

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- Identify and apply design principles such as greediness, divide and conquer and dynamic programming in the design of efficient algorithms
- Describe efficient algorithms for a range of computational problems, along with their computational complexity
- Articulate the key concepts and critically evaluate approaches in a clear and rigorous manner

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Syllabus

Unit 1 – Fundamentals of algorithms and algorithm analysis

In this unit you will learn the basic concepts from logic, sets and functions, and proof techniques that underpin the topics of this module. You will also be introduced to computational problems and how to use and analyse algorithms to solve these problems. You will have the opportunity to hear from an AI practitioner on their perspective on the contents of this module and AI in general. Depending on your background, you may encounter some familiar topics, while others will be new to you. You are encouraged to manage your time spent on this unit flexibly, spending more on the new topics and those that you feel need revision, and less on the familiar ones.

Worksheets are provided for each topic. Use the questions on the slides and in the worksheets to develop and test your understanding. Worksheets are not assessed, but they form the backbone of your learning journey and they prepare you for the courseworks and the final assessment. You should try to answer as many questions as possible, as this is essential for succeeding in this module.

There are also formative assessments at the end of each unit. These assessments do not contribute towards your final grade and are in place to help check your understanding of each unit.

Unit 2 – Graphs and algorithm design principles

In this unit you will consolidate your understanding of asymptotic analysis and explore the stable matching problem. You will then focus on graphs, examining undirected graphs, connectivity, graph representations, breadth first search and depth first search. You'll learn about page rank and the problem of the random surfer. The unit concludes with an introduction of the first basic algorithmic principle for the design of algorithms, namely, greedy algorithms that will be introduced using various classic examples such as Prim's algorithm and Kruskal's algorithm.

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Unit 3 – Algorithm design principles

In this unit you will learn about greedy algorithms for the shortest path problem. You will then examine how to use the divide and conquer approach to solve problems recursively and how to analyse the run-time of recursive algorithms. Finally, you will learn the basics of dynamic programming and how to design algorithms based on dynamic programming.

Unit 4 – Important data structures and introduction to complexity

In this unit you will focus on learning about the most important implementations for the dictionary data structure. In particular, you will learn about search trees, AVL-trees, B-trees and hashtables. Moreover, you will be introduced to the fundamentals of the (time)-complexity of computational problems. Here, you will learn about how to distinguish easy (tractable) from hard (intractable) problems and how to compare and analyse computational problems based on their complexity.

Unit 5 – Reductions and branch and bound

In this unit you will learn about another implementation of the dictionary data structure, which is called hash tables. You will then be introduced to the basics about computational complexity. You will learn about how to distinguish easy (tractable) from hard (intractable) problems and how to compare and analyse computational problems based on their complexity. You will also learn how to use reductions between computational problems to show tractability or intractability. Finally, you will learn about branch-and-bound algorithms, which can be used to solve NP-complete problems to optimality. Moreover, while branch and bound algorithms run in exponential time, they will usually (in practice) be much faster than simple brute-force algorithms and are nowadays by far the most prominent way to solve NP-hard problems to optimality.

Unit 6 – Dealing with NP-hard problems

In this unit you will learn how to obtain good solutions for NP-hard problems quickly. That is, in many cases it is sufficient to obtain a reasonably good solution (i.e., being as close to the

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optimum as possible). To achieve this you will learn about approximation algorithms, i.e., algorithms that run in polynomial-time and allow you to obtain a solution that is guaranteed to be relatively close to the optimum solution. We then show you how to design even quicker algorithms (so called heuristics) that allow you to obtain some solutions quickly; however, they no longer have a guarantee on the quality of the returned solution. Heuristic algorithms also usually have the advantage that they can easily be adapted to a wide variety of problems.

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Key Dates

	Monday	Tuesday	Wednesday	Thursday	Friday
Unit 1	**January 9** Module starts	**10**	**11**	**12**	**13**
Unit 2	**16** Unit 1 webinars	**17**	**18**	**19**	**20**
Unit 3	**23** Assessment 1 release Unit 2 Webinars	**24**	**25**	**26**	**27**
Unit 4	**30** Assessment 1 deadline Unit 3 Webinars	**31**	**February 1**	**2**	**3**
Unit 5	**6** Assessment 2 release Unit 4 Webinars	**7**	**8**	**9**	**10**
Unit 6	**13** Assessment 2 deadline Unit 5 Webinars	**14**	**15**	**16**	**17**
Unit 7	**20** Assessment 3 release Unit 6 webinars	**21**	**22**	**23**	**24**
Study break	**27** Assessment 3 deadline	**28**	**1 March**	**2**	**3**

Please see the module area in Minerva for specific timings for webinars and assessment submission deadlines.

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Module Assessment

You should familiarise yourself with the Digital Education Service Code of Practice on Assessment (CoPA) before you start your first assignment. The CoPA contains important information about marking, late submissions, resits, mitigating circumstances and more.

Assessment 1: Coursework 1

Summary

Multiple choice and free text answer questions

Deadline

The Coursework 1 assignment will be available for you to access for one week, from 08:30am UTC (London) Monday 23rd January.

Due: The deadline for completion and submission is Monday 30th January at 08:30am UTC (London).

Please note: There will be no late submissions for this assignment unless you have been granted an extension

Purpose

The purpose of this assignment is to help you to evaluate your understanding of the algorithms, approaches and principles covered in Units 1 and 2.

Key Information

You will have one week in which to complete the assignment.

The assignment is comprised of 6 questions, with a combination of multiple choice and free text answer questions.

The assignment is accessible in Gradescope. You should record all of your answers in Gradescope.

The points allocation for each question is as indicated on Gradescope. Points will be awarded on an 'all or nothing' basis. This means that for multiple choice questions, you must select all possible correct answers in order to be awarded points.

Academic integrity

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

Submission

Coursework 1 should be submitted via Gradescope.

The link to access the Coursework 1 assignment will be available via Submit my Work once the assessment window is open.

The assignment will be available for one week. During that week, you should work through the questions. You can revisit your assessment multiple times throughout this period and you may change any of your answers until your assignment is submitted. **Please ensure you allow yourself enough time to complete the full assessment for submission by the deadline.**

Please review your answers carefully before submitting your assignment as once you have made the submission no amendments will be possible.

For this assignment type, you **will not** receive email confirmation of your submission.

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Please note: There will be no late submissions for this assignment unless you have been granted an extension.

If you have partially completed the assessment by the deadline, your assessment will be automatically submitted and marked.

If you have mitigating circumstances that prevent you from submitting on time, you may be eligible for an extension, and you should request this before the deadline, providing relevant evidence to support your request.

Marking

This is a compulsory assessment for OCOM5102M Algorithms. It is worth 20% of your overall module grade.

Feedback

In line with normal practice for exams, only the overall mark for the test will be available.

Your grade will be visible via 'Gradebook' on Minerva once marking has been completed. We will notify you by email when your grade is available, usually within 15 working days of the assignment deadline.

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Assessment 2: Coursework 2

Summary

Multiple choice and free text answer questions

Deadline

The Coursework 2 assignment will be available for you to access for one week, from 8:30am UTC (London) Monday 6th February.

Due: The deadline for completion and submission is Monday 13th February 8:30am UTC (London).

Please note: There will be no late submissions for this assignment unless you have been granted an extension.

Purpose

The purpose of this assignment is to help you to evaluate your understanding of the algorithms, design principles and data structures covered in Units 3 and 4.

Key Information

You will have one week in which to complete the assignment.

The assignment is comprised of 6 questions, with a combination of multiple choice and free text answer questions.

The assignment is accessible in Gradescope. You should record all of your answers in Gradescope.

The points allocation for each question is as indicated on Gradescope. Points will be awarded on an 'all or nothing' basis. This means that for multiple choice questions, you must remember to select all possible correct answers in order to be awarded points.

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As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

Submission

Coursework 2 should be submitted via Gradescope.

The link to access the Coursework 2 assignment will be available via Submit my Work once the assessment window is open.

The assignment will be available for one week. During that week, you should work through the questions. You can revisit your assessment multiple times throughout this period and you may change any of your answers until your assignment is submitted. **Please ensure you allow yourself enough time to complete the full assessment for submission by the deadline.**

Please review your answers carefully before submitting your assignment as once you have made the submission no amendments will be possible.

For this assignment type, you **will not** receive email confirmation of your submission.

Please note: There will be no late submissions for this assignment unless you have been granted an extension.

If you have partially completed the assessment by the deadline, your assessment will be automatically submitted and marked.

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If you have mitigating circumstances that prevent you from submitting on time, you may be eligible for an extension, and you should request this before the deadline, providing relevant evidence to support your request.

Marking

This is a compulsory assessment for OCOM5102M Algorithms. It is worth 20% of your overall module grade.

Feedback

In line with normal practice for exams, only the overall mark for the test will be available.

Your grade will be visible via 'Gradebook' on Minerva once marking has been completed. We will notify you by email when your grade is available, usually within 15 working days of the assignment deadline.

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Assessment 3: Final Assessment

Summary

Multiple choice and free text answer questions

Deadline

The Final Assignment will be available to you for one week, from 08:30am UTC (London) Monday 20th February.

Within that week, you will have 8 consecutive hours in which to complete your assignment. You will choose when to start this period of 8 hours and you must ensure that you submit your assignment by the end of this period.

Due: The deadline for completion and submission is Monday 27th February at 08:30am UTC (London).

Purpose

The purpose of this final assessment is to help you to evaluate your understanding of the algorithms, principles and problems covered in the module as a whole. You will need to draw on your knowledge of the topics covered in all six units to be able to complete the assessment.

Key Information

You will have 8 hours in which to complete the assignment.

The assignment is comprised of a combination of multiple choice, file upload and free text answer questions.

The assignment is accessible in Gradescope. You should record all of your answers in Gradescope.

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The points allocation for each question is indicated on Gradescope. Points are awarded on an 'all or nothing' basis. This means that for multiple choice questions in particular, you must select all possible correct answers in order to be awarded points.

For written/file upload questions, you will be awarded points on how convincing, thorough, and well-presented your arguments/algorithms are.

Academic integrity

As a student at the University of Leeds, you need to adhere to the University's regulations on academic misconduct. All work you submit for assessment is expected to be your own independent work. Where you have worked with others or where you present other people's work as part of your assessment, you must fully acknowledge this.

All students should ensure that they are aware of the principles of Academic Integrity. For more information or if you are unsure about academic integrity, please read the University of Leeds' guidance on Academic Integrity. Please note that by submitting this piece of work you are agreeing to the University's Declaration of Academic Integrity.

Submission

Assessment 3 should be submitted via Gradescope.

The link to access the Final Assessment assignment will be available via Submit my Work once the assessment window is open.

The assignment will be available for one week. In that week, you have 8 hours in which to complete and submit your assignment. For example, if you begin the assessment at 10am on Friday 24th February, you must complete it by 6pm that day. **Please note the time will begin as soon as the assessment is accessed, so only start when you are ready.**

Your answers for this assignment will be automatically saved as you go through and once you have completed you will need to submit them.

During the submission period, you can submit any number of times. You will be graded based on your final submission.

The timer will continue counting down even if you leave the Gradescope site.

Once you have submitted, you will see a review screen where the questions and your given answers will appear on the left hand side. The points value that appear against each question are the marks available, and do not reflect your actual score.

For this assignment type, you will not receive email confirmation of your submission.

If you have partially completed the assessment by the deadline, your assessment will be automatically submitted and marked.

Marking

This is a compulsory assessment for OCOM5102M Algorithms. It is worth 60% of your overall module grade.

Feedback

Your grade will be visible via 'Gradebook' on Minerva once marking and moderation has been completed. We will notify you by email when your feedback is available. You will usually receive feedback within 15 working days of the assignment deadline.

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Reading List

The reading list and online course readings for this module can be found in the 'Reading List' section of Minerva.

Additional Information and Support

Key Documents

Please ensure that you read and familiarise yourself with the following important documents, available in the Sustainable Business Leadership Online organisation area on Minerva:

- Digital Education Service (DES) Code of Practice on Assessment
- Code of Practice on Attendance
- Artificial Intelligence Programme Structure

Skills@Library

Skills@Library provides support for students to develop academic skills. This includes areas such as researching, writing, note making and referencing. This support is delivered through online resources, workshops and webinars: http://library.leeds.ac.uk/skills.

General Module Forum

The Digital Education Service and the University of Leeds are committed to supporting students. If you have any questions about the academic content of the module, please post them in the module's 'General Module Forum' discussion board. You can access this through the 'Discussion Board' section of Minerva. The Module Team will reply to your post and your peers will also have access to the answer.

Student Success Advisor

If you have any questions about using Minerva, administrative processes, or pastoral care please contact your Student Success Advisor. They will be able to advise you or signpost you to the correct resource.

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Recording Policy

As part of your study on this module, webinars will be recorded by the University. This recording is conducted in accordance with the University's Policy on Audio or Video Recording for Educational Purposes. Please read the Policy for further background information relating to your rights. The purpose of the recording is to support your study at the University. It will be accessible by students enrolled on the module through Minerva.

In accordance with the University's Intellectual Property Policy in general where staff or students make a recording, all rights in that recording (including rights in the sounds made) will belong to the University. The University seeks to respect student and staff rights to be acknowledged as authors and performers. Otherwise, to the extent allowed under the general law, any individual interests in the recordings are waived, allowing the University to act in effect as a custodian, to maximise educational objectives in the general interest of all.

If you have any concerns relating to the recording please contact the individual leading the educational activity, or the module leader. If you wish to opt out of being recorded or wish to request that your contribution be edited please contact the individual leading the activity or the module leader in advance of the session (for optouts) or immediately after the session (for requests for edits). Please note that you may not opt out of recordings which form part of the assessment process on the module. If you have any concerns relation to the recording, please contact your Student Success Advisor.

Complaints

If you have any complaints about this module, please write to DESStudents@leeds.ac.uk and the Digital Education Service will review your complaint. If a remedy cannot be arrived at, the University's official complaints procedure could be followed.

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OCOM5102M Algorithms

Unit 1: Fundamentals of algorithms and algorithm analysis

Lesson 1.1: Mathematical foundations: logic
Propositional logic
Predicate logic

Lesson 1.2: Mathematical foundations: sets and relations
Set, set equality, cardinality
Subsets and supersets, proofs involving subsets
Set operations and proofs
Pairs, tuples and Cartesian products
Relations: definition, examples, properties
Equivalence relations, order relations

Lesson 1.3: Mathematical foundations: proof techniques
Functions
Proof

Lesson 1.4: Example stable matching
Stable matching: the problem
The Gale-Shapley algorithm
GS algorithm: discussion

Lesson 1.5: People in AI
Dr Siamak Tazari, Google

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Unit 2: Analysis of algorithms, graphs, and greedy algorithms

Lesson 2.1: Analysis of algorithms I
Analysis of algorithms: introduction
Analysis of algorithms: pseudocode
Analysis of algorithms: asymptotic analysis (introduction)

Lesson 2.2: Analysis of algorithms II
Analysis of algorithms: asymptotic analysis (further properties)
Analysis of algorithms: asymptotic analysis (common functions)
Analysis of algorithms: asymptotic analysis (common runtimes)
Analysis of algorithms: asymptotic analysis (stable matching)

Lesson 2.3: Graphs I
Graphs: basic definitions and applications
Connectivity: undirected graphs
Graph representations (data structures)

Lesson 2.4: Graphs II
Graph traversal (breadth first search)
Graph traversal (depth first search)
Connected components
Connectivity in directed graphs
DAGs and topological sort

Lesson 2.5: People in AI
Professor Hannah Bast, University of Freiburg

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Unit 3: Greedy, divide and conquer, and dynamic programming algorithms

Lesson 3.1: Page rank
Page rank: introduction
Page rank
Page rank: the random surfer
Page rank random surfer and page rank property: proof of theorem
Page rank: Markov chains

Lesson 3.2: Greedy algorithms
Greedy algorithms
Greedy algorithms: minimum spanning tree problem
Greedy algorithms: Prim’s algorithm
Greedy algorithms: Kruskal’s algorithm
Correctedness of Prim’s and Kirskal’s algorithms

Lesson 3.3: Greedy algorithms
Greedy algorithms: Dijkstra’s algorithm
A-star search algorithm

Lesson 3.4: Divide and conquer I
Divide and conquer: introduction
Divide and conquer: sorting (introduction)
Divide and conquer: sorting (mergesort)
Divide and conquer: sorting (quicksort)
Divide and conquer: sorting (comparison)

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Lesson 3.5: People in AI
Molham Aref, Relational AI
Dr Hung Ngo, Relational AI

Unit 4: Dictionary data structures and computational complexity

Lesson 4.1: Divide and conquer II
Divide and conquer: recurrences (introduction)
Divide and conquer: recurrences (guessing a solution)
Divide and conquer: recurrences (master method)
Divide and conquer: closest pair of points

Lesson 4.2: Dynamic programming I
Dynamic programming: introduction
Dynamic programming: weighted interval schedulingExample topic
Dynamic programming: segmented least squares Example topic

Lesson 4.3: Dynamic programming II
Dynamic programming: shortest path (overview)Example topic
Dynamic programming: Bellman-Ford algorithm
Dynamic programming: BF algorithm (improvements)
Dynamic programming: matrix multiplication

Lesson 4.4 Dictionary problem and search trees I
Dictionaries: introduction
Search trees
Tree traversals

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Binary search trees

Lesson 4.5 Dictionary problem and search trees II
AVL trees

Lesson 4.6: People in AI
Professor Moshe Vardi, Rice University

Unit 5: Reductions and branch and bound

Lesson 5.1: Hash tables
Hash tables: introduction
Hash tables: hash functions
Hash tables: chaining
Hash tables: open hashing

Lesson 5.2: Tractability and polynomial time reductions
Introduction: (in-)tractability
Polynomial time reductions
Polynomial time reductions: applications
Polynomial time reductions: Boolean satisfiability
Decision vs. finding vs. optimisation

Lesson 5.3: Complexity classes and polynomial time reductions
The complexity class NP
NP-completeness
Further examples of reductions: constraint satisfaction problems

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Further examples of reductions: sequencing problems
Further examples of reductions: partitioning problems

Lesson 5.4: Branch and bound algorithms I
Branch and bound: introduction
Knapsack: introduction
Knapsack: brute-force

Lesson 5.5: Branch and bound algorithms II
Knapsack: branch and bound

Lesson 5.6: People in AI
Professor Martin Dyer, University of Leeds
Dr Wisdom Agboh, University of Leeds

Unit 6: Dealing with NP-hard problems using approximation algorithms

Lesson 6.1: Extending tractability
Extending tractability: introduction
Knapsack

Lesson 6.2: Approximation algorithms I
Approximation algorithms: introduction
Approximation algorithms: vertex cover
Approximation algorithms: LP rounding - weighted vertex cover

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Lesson 6.3: Approximation algorithms II
Approximation algorithms: centre selection
Approximation algorithms: 1/2-approximation for knapsack
Approximation algorithms: PTAS for knapsack

Lesson 6.4: People in AI
Sian Carey, University of Leeds
Lucy Godson, University of Leeds
Sarah Smith, University of Leeds

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